we know by the construction in Theorem 29.3 that $\mathbb{C} \simeq \mathbb{R}[x]/\langle p(x)\rangle$ for any irreducible polynomial $p(x)$ in $\mathbb{R}[x]$ of degree 2.

Now let E be any finite extension of $\mathbb{R}$. If $E \neq \mathbb{R}$, then let β be in E but not in $\mathbb{R}$. Then $p(x) = \text{irr}(\beta, \mathbb{R})$ has degree 2,because we have seen that there are no irreducible polynomials in $\mathbb{R}[x]$ of greater degree. The construction in Theorem 29.3 shows that $\mathbb{R}(\beta) \simeq \mathbb{R}[x]/\langle p(x)\rangle$ and hence $\mathbb{R}(\beta) \simeq \mathbb{C}$. Because $\mathbb{C}$ is algebraically closed, $\mathbb{R}(\beta)$ is algebraically closed also and admits no proper algebraic extensions. Because E is an algebraic extension of $\mathbb{R}(\beta)$, we must have $E = \mathbb{R}(\beta)$, so $E \simeq \mathbb{C}$.

38. If R contains no nontrivial proper ideals, then $\{0\}$ is the only proper ideal, and is a maximal ideal, and of course it is contained in itself so we are done.

Suppose R contains a nontrivial proper ideal N which of course does not contain the unity 1 of R. The set S of ideals of R that do not contain 1 is partially ordered by inclusion. Let $T = \{N_i \mid i \in I\}$ be a chain of S. We claim that $U = \bigcup_{i\in I} N_i$ is an element of S that is an upper bound of T. Let $x, y \in U$. Then $x \in N_j$ and $y \in N_k$ for some $j, k \in I$. Because T is a chain, one of these ideals is contained in the other, say $N_j \subseteq N_k$. Then $x, y \in N_k$ which is an ideal, so for all $r \in R$, we see that $x \pm y, 0, rx$, and xr are all in N_k and hence in U. This shows that U is an ideal. Clearly $N_i \subseteq U$ for all $i \in I$, and 1 is not in U because 1 is not in N_i for any $i \in I$. Thus $U \in S$ and is an upper bound for T, so the hypotheses of Zorn's lemma are satisfied.

Let M be a maximal element of S; such an element of S exists by Zorn's lemma. Because $M \in S$, we see that M is an ideal of R, and does not contain 1 so $M \neq R$. Suppose that L is an ideal of R such that $M \subseteq L \subseteq R$. If $L \in S$, then $M = L$ because M is a maximal element of S under set inclusion. Otherwise, $1 \in L$ so $L = R$. Thus M is a maximal ideal of R.

32. Geometric Constructions

1. By Euler's formula, $e^{i(3\theta)} = \cos 3\theta + i \sin 3\theta$. On the other hand, $e^{3i\theta} = (e^{i\theta})^3 = (\cos\theta + i\sin\theta)^3$, which we compute using the binomial theorem; the real part of the answer should be equal to $\cos 3\theta$. We have

$$(\cos\theta + i\sin\theta)^3 = (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(\text{don't care}).$$

Thus $\cos 3\theta = \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

2. T T T F T F T T T F

3. If a regular 9-gon could be constructed, the angle $(360/9)^\circ = 40^\circ$ could be constructed, and could then be bisected to construct an angle of 20°. The proof of Theorem 32.11 shows, however, that an angle of 20° is not constructible.

4. One can construct an angle of 30° if and only if one can construct the number $\cos 30^\circ = \sqrt{3}/2$. Because $\sqrt{3}$ is constructible and quotients of constructible numbers are constructible, an angle of 30° is constructible.

5. Because $|\overline{OA}| = |\overline{OP}|, \angle OAP = \angle APO = (180^\circ - 36^\circ)/2 = 72^\circ$. Then $\angle QAP = 36^\circ$ so triangle OAP is similar to triangle APQ. Now $|\overline{AP}| = |\overline{AQ}| = |\overline{OQ}| = r$, so $|\overline{QP}| = 1 - r$. Taking ratios of corresponding sides, we obtain $|\overline{AP}|/|\overline{OP}| = |\overline{OA}|/|\overline{AP}|$ so $\frac{r}{1-r} = \frac{1}{r}$. Thus $r^2 = 1 - r$ so $r^2 + r - 1 = 0$. By the quadratic formula, we find that $r = \frac{-1+\sqrt{5}}{2}$ which is a constructible number. Thus we can construct an angle of 36° by taking a line segment $\overline{OP}$ of length 1, drawing a circle of radius 1 at O and one of radius r at P, and finding a point A of intersection of the two circles. Then $\angle AOP$ measures